

AUTHENTICLENS

KNOWLEDGE BASE - DOCUMENT 9 OF 9

Document 9 - Apparel, Product, Luxury Styling, and Collage Production QA

Deep Fashion/Product Protocols for Melato-Style E-Commerce and Realistic Composite Design

Purpose: upload-ready operational documentation for the AuthenticLens Gemini Gem. This document translates theory into retrieval-friendly procedures, decision rules, prompt modules, and quality-control checks.

Prepared for: AuthenticLens - The Realism Architect

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Language: English (Canada)

Primary target: Gemini Pro + Nano Banana / Nano Banana Pro image generation and editing workflows

Purpose

Provide a deeper operational layer for apparel, fashion, product photography, clothing swaps, luxury styling, group collages, and e-commerce image sets. Document 5 gives the general enhancement framework; Document 9 gives production-specific controls and acceptance standards.

Intended Gemini Behaviour

- Make apparel look physically manufactured, not painted or hallucinated.
- Preserve product design, colour, panels, stitching, and material identity.
- Produce consistent e-commerce image sets for product pages.
- Handle luxury styling without garbled logos or counterfeit-looking pseudo-text.
- Blend group collages without changing faces or body identity.

A. Apparel Material Physics

Material	Realism cues	Common AI/edit failure
Velour	Directional nap, soft crushed highlights, pile texture, rich but not plastic colour.	Uniform velvet-like blur, glowing highlights, no weave/nap direction.
Fleece / sweatshirt knit	Soft loft, rib cuffs, seam thickness, small lint/fibre variation.	Flat painted fabric, no cuff structure.
Wool coat	Weighty drape, structured shoulders, lapel thickness, subtle fuzz.	Rubber-like smooth coat, weightless folds.
Denim	Twill weave, seam bulk, whiskers, knees/seat creases.	Random noise instead of weave; impossible seam paths.
Leather	Grain variation, edge wear, crease memory, specular highlights.	Plastic shine or perfectly uniform surface.
Nylon/puffer	Panel quilting, seam pressure, subtle reflections, inflated volume.	Melted baffles, floating zipper, impossible puff symmetry.

B. Garment Construction Checklist

- Shoulder seams sit on plausible shoulder points.
- Collars have thickness and cast small shadows on the neck/chest.
- Zippers follow one continuous track with consistent tooth spacing.
- Pockets have openings, seam borders, and plausible depth.
- Cuffs/ribbing wrap around wrists/ankles without melting into skin.
- Pants drape from waist to hips to knees to hem according to fabric weight.
- Logos/patches follow garment curvature and perspective.

C. Melato-Style Product Page Shot System

Shot	Purpose	Required realism
Hero front	Instant product recognition.	Full garment visible, sharp, accurate colour, natural shadow.
Back view	Shows back motif/paneling.	Same colour and scale as front shot.
3/4 model	Shows fit and drape.	Natural body posture, real folds at joints.
Side view	Shows silhouette and garment thickness.	Correct sleeve/pant length and seam continuity.
Fabric macro	Proves material quality.	Actual textile structure, nap/weave, no fake noise overlay.
Zipper/hardware macro	Shows premium construction.	Clean metal reflection, teeth, pull tab, stitching.
Patch/logo detail	Shows branding/design story.	Legible, aligned, sewn-on texture, no pseudo-text.
Lifestyle/hero scene	Brand mood for homepage or lookbook.	Realistic environment, coherent lighting, no identity drift.

D. Luxury Outfit Swap Protocol

Default styling rule when the user requests premium clothing but does not name a brand: use restrained Saint Laurent-style language - minimal tailoring, clean dark palette, sharp coat or shirt, understated luxury. Avoid forcing large logos.

1. Lock identity: face, complexion, body, hands, pose, and expression remain unchanged.
2. Describe garment silhouette: coat, shirt, trousers, jacket, tracksuit, or accessory.
3. Describe material: wool, cotton poplin, leather, velour, denim, nylon, etc.
4. Describe construction: lapels, seams, zipper, cuffs, hem, fit, pockets.
5. Match light: garment highlights and shadows must match original subject lighting.
6. Avoid unreadable logos. If brand mark is requested, use minimal, readable placement or no visible mark.
7. Check body fit: sleeves align with wrists, collar around neck, hem falls naturally.

E. Logo and Text Fidelity

Scenario	Instruction
Brand not required	Avoid visible logos; use style/material cues instead.
Brand requested	Use clean, readable typography only if the model can render it accurately. Otherwise use understated logo-free styling.
Product patch	Patch must follow fabric curve, have stitch border, proper shadow, and readable graphic.
Background signage	Use generic blurred signage unless exact readable

	signs are verified and necessary.
E-commerce text overlays	Prefer adding text outside the generated image in design software rather than asking the image model to render complex text.

F. Group Collage Production Workflow

8. Extract subject boundaries carefully: preserve hair edges and shoulders without cutting into face/body.
9. Normalize scale: align eyes/head height based on intended depth position.
10. Normalize colour: match white balance and contrast before compositing.
11. Normalize grain: add a unifying mild grain/noise pass after all subjects are placed.
12. Build shared lighting: add consistent shadow side, rim light, and contact shadow.
13. Use background depth: soft gradients, realistic studio or street backgrounds, and foreground/background blur help integration.
14. Final face check: compare each face to source; reject if identity drift is visible.

G. Acceptance Standards for Product and Fashion Images

Dimension	Pass standard
Identity	Subject remains recognizable; no facial reshaping or complexion drift.
Product fidelity	Design, colourway, panels, logos/patches, seams, and garment type preserved.
Material realism	Fabric type is visually believable at normal and zoomed view.
Construction	Seams, zippers, cuffs, hems, collars, pockets, and patches make physical sense.
Lighting	Highlights, shadows, and contact shadows are coherent.
E-commerce quality	Sharp, clean, colour-accurate, consistent crop, no distracting artifacts.
Brand safety	No garbled logos, fake unreadable typography, or counterfeit-looking text.

H. Prompt Modules

Goal	Prompt module
Velour tracksuit enhancement	Enhance the product realism while preserving the exact tracksuit design and colourway. Make velour nap, stitching, zipper hardware, rib cuffs, seams, and fabric weight visibly premium. Keep natural shadows and avoid plastic shine.
Product macro	Create a realistic macro detail shot of the existing product fabric and construction. Show textile weave/nap, stitch spacing, zipper teeth, and

	material depth. Do not invent new logos, colours, or pattern details.
Saint Laurent-style coat swap	Change only the clothing to a minimal Saint Laurent-style tailored coat with premium dark wool, clean lapels, realistic thickness, and natural drape. Preserve face, body, complexion, pose, and lighting.
Collage blend	Blend the provided subjects into a cohesive design without altering faces. Match scale, colour grade, grain, shadows, and edge softness. Use a smooth professional background that makes them feel part of one composition.
Melato e-commerce set	Generate a consistent product-photo set: front, back, 3/4, side, fabric macro, zipper macro, patch/logo detail, lifestyle shot. Maintain exact design, colour consistency, premium clarity, and realistic garment construction.

Quality-Control Procedure

- Does the product still match the original design?
- Are fabric, stitching, and zipper details physically plausible?
- Are logos/patches readable and correctly placed?
- Does the garment drape on the body naturally?
- Do all images in a set share colour, light, and material consistency?
- For collages: did any face change?
- Would this image look credible on melato.ca without exposing AI artifacts?

Quick Reference Summary

Fashion realism is construction realism. Product quality depends on fabric detail, colour accuracy, seams, stitching, hardware, fit, and consistent light. Avoid fake logos and identity drift. For e-commerce, preserve design over creative invention.

Source Notes and Reliability Labels

Use source notes as operational grounding, not as a claim of legal or forensic certainty. If metadata, original files, or verified provenance are unavailable, AuthenticLens must label final authenticity claims as probabilistic.

- **Internal Doc 1:** AuthenticLens_Doc1_AI_Artifact_Lexicon.pdf - primary artifact taxonomy and detection guidance.
- **Internal Doc 2:** AuthenticLens_Doc2_Photographic_Physics.pdf - optical physics, sensor/film artifacts, lighting, exposure, and colour foundations.

- **Internal Doc 3:** AuthenticLens_Doc3_Realism_Prompting_Formulas.pdf - realism prompt formulas, camera/film language, and platform prompting references.
- **Google AI Developers:** Image generation with Gemini / Nano Banana and Nano Banana Pro documentation. Source note: confirms Gemini image generation/editing is conversational, supports text+image inputs, identifies Nano Banana as gemini-2.5-flash-image and Nano Banana Pro as gemini-3-pro-image-preview, and states generated images include SynthID watermarking.
- **Adobe Lightroom Help:** Lightroom documentation for Texture, Clarity, Dehaze, Vignette, Grain, Sharpening, Noise Reduction, local adjustments, and masking. Source note: used to separate editing artifacts from AI-generation artifacts.
- **C2PA Specification:** C2PA Content Credentials specification and explainer. Source note: provenance is represented by signed manifests / Content Credentials that can record creation and modification assertions.
- **NIST Digital Media / Deepfake Guidance:** NIST digital and multimedia evidence resources and deepfake guidance. Source note: provenance, supply chain, compression, and context matter; visual inspection alone is not definitive.
- **W3C sRGB:** sRGB as the default colour space for web images. Source note: used for e-commerce and social-media colour delivery guidance.

Reliability labels used in this knowledge base: Confirmed source, Common photography practice, Image-forensics heuristic, Operational guidance, Prompting heuristic, Model-behaviour observation.